TYREMIND

Causal tyre intelligence · Technical dossier

Every published tyre model is graded on how well it predicts lap times. None of them checks whether the **reason** it gives for those lap times is the right one. On 3 of the 4 real 2024 races analysed, the standard method reports that tyres get **faster** the longer they run.

TyreMind is built around that gap. It characterises why separating tyre wear from fuel burn, track evolution and traffic is structurally hard — three collinearities, two of which no amount of data can resolve — builds an estimator that handles them, and then tests the attribution against a truth it controls. Recovering a hidden degradation rate, its error is **0.0044 s/lap** against the standard method's **0.0966**.

95.5%
Error reduction recovering a known rate
25 synthetic sessions, 75 comparisons

0.22 ms
Per-lap update, live
flat in session length, 921 laps

7 / 8
Circuits whose rotation the physics recovered
never told the answer

22.7
RUL RMSE on NASA turbofans, same estimator
all 100 test engines, no tyre-specific code

Challenge	TrackShift Innovation Challenge — Problem 3, AI Motorsport Intelligence
Brief	Isolate true tyre wear rates from confounding practice variables; validate predicted wear against actual race-day pace
Repository	`github.com/nitya-prakash-pandey-2005/TyreMind`
Evidence	203 sessions · 92,326 laps · 84 event-seasons across 2022–2025
Scale	20,354 lines of Python · 7,049 lines of TypeScript · 326 tests · 24 API endpoints · 18 recorded experiments
Status	Runs fully offline. Every figure in this document is read from a committed result file, not typed by hand.

This document states what TyreMind does not do as prominently as what it does. A section on limitations, failure modes and the competitor that beats us appears before the commercial case, not after it.

00 Contents

How to read this document in ninety seconds

Section **03** is the intellectual core: the algebra showing why this problem is hard in a way the brief does not say. Section **08** is the evidence. Section **10** is the one-page answer to "what is actually new here". Section **11** is where we say what is wrong with it, including a competitor that beats us on the metric everyone else reports.

A note on numbers

Every quantity in this dossier carries a provenance stamp naming the file it came from. Those files are committed to the repository and regenerated by running the experiment scripts. Nothing here was rounded into a better story; where a result is unflattering — a failed hypothesis, a systematic bias, a model that beats ours — it appears with the same weight as the results that help.

That is not modesty. Tyre degradation is used to make pit-stop decisions worth championship points, and a tool that cannot state its own error bars has no business in that room.

01 The problem, stated precisely

Why "watch the lap times climb" is not merely imprecise but wrong-signed

A Formula 1 team needs one number to decide when to pit: how much slower the car gets per lap because of the tyre. The obvious way to measure it is to fit a line through lap time against tyre age over a stint.

That method does not merely add noise. It returns the wrong sign. Over a stint the car burns roughly 2.7 kg of fuel per lap, and each kilogram is worth about 0.030 s of lap time, so the car gets *faster* by around 0.081 s/lap from fuel alone. Meanwhile the circuit itself gains grip as rubber goes down, worth close to a second over a session. Both effects move lap time smoothly with lap number — exactly like degradation, and in the opposite direction, and often larger.

-0.004	-0.034	0.074	0.113
Naive degradation, HARD, Monza 2024 race	Naive degradation, MEDIUM, same race	TyreMind, HARD, same laps	TyreMind, MEDIUM, same laps
s/lap — tyre getting faster	s/lap — also impossible	±0.018 s/lap	±0.023 s/lap

`source live API, /api/session/2024-monza-R · 921 laps, 20 cars`

A negative degradation rate is not a subtle miscalibration that a practitioner can mentally correct. It is a claim that tyres improve with use, and it is what the standard method actually reports on real Grand Prix data. The confounders are not a nuisance term around the signal; on a race stint they are *larger* than the signal.

What the brief asks, and what that implies

The brief asks to "strip out external noise from practice sessions to generate clean tyre performance degradation curves", and to validate predicted wear against actual race-day pace. Read carefully, that is not a prediction problem. It is an **attribution** problem: of the observed change in lap time, how much was the tyre?

> **The distinction that organises everything that follows**
>
> Fuel burn-off, track evolution and tyre degradation all move lap time smoothly with lap number. **Many wrong decompositions sum to the same right total.** A model can predict lap times almost perfectly while attributing the change to entirely the wrong cause — and no lap-time metric will notice, because the total is right.
>
> Every model we surveyed is validated on lap-time prediction error. So the failure that matters in this brief is precisely the failure the field's standard metric cannot see.

Who needs the answer, and what changes when they have it

Decision	Made today with	What a clean degradation rate changes
Pit lap	A fitted slope from Friday practice, adjusted by engineering judgement and a fuel correction of unstated confidence	The slope arrives with an interval whose width came from a stated identifiability argument, so the strategy simulation can propagate it instead of assuming it away
One stop or two	Compound deltas from practice, often statistically indistinguishable when fitted per driver	Pooling the whole field separates compounds that a single-car fit cannot
Undercut or hold	Gap and pace deltas, tyre state inferred by eye	Live per-car degradation updating every lap, with the interval collapsing visibly as evidence

Decision	Made today with	What a clean degradation rate changes
		accumulates
Set allocation across a weekend	Historical rules of thumb per circuit	Per-corner energy from the circuit's own racing line, which explains why the same compound behaves differently between circuits

source docs/DEMO_STORY.md · docs/model_card.md §intended use

02 What already exists, and what it leaves open

Surveyed before implementation, so that rebuilding someone else's result could be ruled out deliberately

"AI predicts tyre degradation" describes work that is published, peer-reviewed and in some cases running live at Grands Prix. The first output of this project was a research audit written *before* any modelling code, whose purpose was to find what was genuinely unaddressed rather than to justify a plan already made.

Work	What it does	What it leaves open
Cappello & Hoegh arXiv:2512.00640, Nov 2025	Bayesian state-space model of F1 tyre degradation. Lap time as a function of fuel mass and latent tyre pace; pit stops as state resets; compound-specific rates; skewed-*t* observations; FastF1 data.	**Single driver, single race.** No traffic, no track evolution, no telemetry, no physics. Reports compounds as statistically indistinguishable — a statistical power problem that pooling across the field addresses directly.
Pitwall arXiv:2607.06495, 2026	Calibrated Monte Carlo race simulator (2,000 sims, fitted on 191k green laps) plus an LLM briefing layer gated on verified claims. Ran live at the 2026 Austrian and British Grands Prix.	Own stated gaps: **no car telemetry**, fuel mass approximate, tyre thermals absorbed into noise. Predicts race *outcomes*; does not attribute observed pace to causes.
Explainable tyre energy arXiv:2501.04067	XGBoost and deep models predicting tyre energy, with SHAP attributions and counterfactual explanations.	Uses **private Mercedes-AMG telemetry**. Not reproducible from public data by anyone outside the team.
MegaRide / thermodynamic wear S0043164824000565	Tyre wear model fusing rubber viscoelasticity, road roughness and thermodynamics. State of the art in physical fidelity.	Commercial; requires rig and finite-element data that public sources do not contain.
Heilmeier et al. TUMFTM/race-simulation	Open-source race simulator and pit-stop optimisation. The reference implementation in the field.	Takes a degradation rate as **given**. The number this project produces is that simulator's most important input.
Rubber wear review PMC12915245	Survey of tyre wear modelling. Archard as baseline; energy-based formulations as the better-supported family.	Confirms that a constant Archard coefficient is inadequate for rubber and that temperature belongs inside the coefficient — which shaped our wear model.
Multilevel F1 performance models Bristol / White Rose, 1950–2014	Hierarchical modelling of driver and constructor contributions to race outcome.	Establishes pooling in motorsport, but not applied to tyre degradation, and does not use pit stagger as identifying variation.
NASA C-MAPSS PCoE repository	Turbofan run-to-failure benchmark with published remaining-life labels; a large comparable literature.	Not motorsport — which is exactly why we use it: it supplies the ground truth F1 cannot.

source `docs/research/01_RESEARCH_AUDIT.md §1`

> **Conclusion of the audit**
>
> Degradation prediction and race simulation are solved-enough problems. Rebuilding either was ruled out. What no surveyed work does is test whether the decomposition it reports is *correct* — and that is the thing the brief actually asks for.

Why the gap has survived

Not because the field is careless. The gap exists because testing attribution requires a ground truth that motor-sport cannot supply. Public F1 data contains no measured tread depth, no rubber mass, no wear sensor. There is no column to score an attribution against, so the field scores what it can measure: the lap time. The metric is not wrong, it is simply blind to the failure mode that matters here.

TyreMind's response is to manufacture the missing ground truth in two directions at once. **Synthetically**, by generating sessions where the true degradation rate is set by us and buried under realistic confounding, so recovery can be measured per cause. And **cross-domain**, on NASA's turbofan benchmark, where run-to-failure labels do exist and the identical estimator can be scored against them.

03 Identifiability: the actual hard part

Three collinearities. Only the first is in the brief. Two cannot be resolved by any amount of data.

Working through the algebra before building revealed that this problem has a specific structural difficulty, and that most of it is invisible until you write the equations down. What follows is the derivation, the resolution chosen for each case, and — stated plainly — whether that resolution came from evidence or from assumption.

3.1 Fuel against degradation, within a run

Both are linear in laps completed. Fuel makes the car faster at roughly 0.030 s/kg × 2.7 kg/lap ≈ **0.081 s/lap**; the tyre makes it slower by an unknown amount. Within one run these are not separable at all — not poorly separable, *not separable*.

The unknown *starting* fuel mass is irrelevant, because it shifts the run's intercept and a run-level effect absorbs it. Only the slope matters, and the slope is exactly confounded.

> **Resolution — by assumption**
>
> Pin the fuel coefficient with an informative physical prior (0.030 ± 0.005 s/kg; 2.7 ± 0.3 kg/lap) and carry its uncertainty as state, so it *widens every published interval* rather than being silently assumed away.

Verified empirically. On 25 synthetic sessions the naive estimator's bias is −0.0966 s/lap — almost exactly the fuel slope, in the direction theory predicts. The collinearity shows up as a measured quantity, not as an argument.

The algebra, in full

“Not separable” is a strong claim and deserves the two lines that establish it rather than an assertion. Within one run, lap time under the structural model is

$$y_t = c_r - \varphi f_t + \beta a_t + \varepsilon_t$$

with f_t the laps completed on the run (fuel burned) and a_t the tyre's age. A run puts exactly one lap on the tyre per lap completed, so $a_t = a_0 + f_t$, and substituting gives

$$y_t = [\, c_r + \beta a_0 \,] + (\beta - \varphi) f_t + \varepsilon_t$$

The run intercept c_r is free, so it swallows βa_0 whole. What remains is a single slope multiplying a single regressor. **Only the combination $(\beta - \varphi)$ is identified.** The Fisher information matrix for the pair is singular by construction, and its determinant is zero rather than small.

> **This rules out the obvious rescue**
>
> Pit-stop stagger does *not* fix it. Different cars starting a stint on tyres of different ages varies a_0, and a_0 is absorbed by the run intercept. More cars does not help. More laps does not help. A longer season does not help. The whole-field pooling that identifies *other* parts of this model contributes nothing here.

Measured, not argued

Experiment 18 checks the algebra against the data rather than trusting it. Across 520 runs in 11 sessions, the within-run correlation between the two regressors is **1.000000000** and the Fisher information is rank-deficient in **11 of 11** sessions. With a flat prior on φ the posterior variance of β diverges in every one of them.

This got *stronger* during development, which is worth recording. An earlier corpus showed ~95% exact collinearity and two ranks of two, because the fuel regressor was a positional counter over rows surviving the quality filter rather than laps actually completed. The residual identification was a **filtering artefact**, not physics. Correcting the counter removed it and made the result exact — a fix that made the project's own position harder to defend, which is the direction a real correction usually runs.

So what buys it back?

Two channels, and there are only two.

The prior. With $\varphi \sim N(\mu, \tau^2)$, the posterior precision of β is finite; let $\tau \to \infty$ and it diverges. The prior is not a refinement of this estimate, it is the reason the estimate exists. The Cramér–Rao bound it implies is **0.0161 s/lap**, against the model's own reported **0.0221** — the estimator sits **1.37×** off the theoretical floor, which is a tighter efficiency check than anything else in this document.

Curvature. Fuel burns at a constant rate, so its effect is exactly linear in laps. The tyre's is not: the degradation rate is carried as a random walk, so accumulated loss may bend, and any departure from a straight line is orthogonal to the fuel term and identified by data alone. Measured, that channel is **7.8%** of the tyre term (median across sessions, range 0.6% to 49.7%).

> **The uncomfortable corollary**
>
> The flatter the true degradation curve, the more the answer rests on the prior. A perfectly linear tyre is one whose degradation cannot be separated from fuel by any quantity of data, at any circuit, in any season. Every degradation figure in this document should be read as conditional on 0.030 s/kg — and section 11 quantifies how far the answer moves when that assumption is wrong by a full standard deviation.

3.2 Track evolution against a uniform shift in degradation

Not anticipated. This one was found by diagnosing a stubborn −0.013 s/lap bias that survived removing the cliff, scrubbed sets and traffic from the generator — which is to say, it was found the hard way.

Shift every degradation rate by c and the track-evolution slope by $-c$. For a lap at session lap L with tyre age $a = a_0 + (L - L_0)$:

```
degradation moves by  +c·(a₀ + L − L₀)
track       moves by  −c·L
─────────────────────────────────────
net                    c·(a₀ − L₀)    ← constant WITHIN a run
```

A constant within a run is exactly what the run intercept absorbs. The two are structurally indistinguishable, and — this is the part that took a day to accept — scrubbed sets do not help, because the difference is still constant within the run.

> **Resolution — by assumption, but a physically correct one**
>
> Model track evolution parametrically as a saturating curve `A·(1 − exp(−kL))` with an informative prior on the amplitude, rather than as a free random walk. Rubber deposition genuinely saturates, so this is the more physically correct model as well as the identifying one. **Bias fell from −0.013 to +0.0012 s/lap.**

3.3 Tyre age against session lap, within a run

They advance together. A single car cannot distinguish an ageing tyre from a changing session — the two variables are the same variable plus a constant.

> **Resolution — by evidence, and it is free**
>
> Fit the whole field at once. Cars change tyres on different laps, so at any session lap the grid spans a wide range of tyre ages. **Pit stagger is a natural experiment** requiring no prior — only the whole grid instead of one car. This is the structural difference from Cappello & Hoegh, who model one driver, and it is why they find compounds statistically indistinguishable where we separate them.

What this costs, quantified

Two of three are resolved by assumption. That is a real weakness and it is measured rather than described: experiment 02 re-fits everything under perturbed priors and reports how far the answer moves.

Variant	What it asks	Rate MAE	Bias	Mean sd	Coverage
baseline	Priors as documented	0.0047	+0.0043	0.017	100%
live at 75% of the session	Forward-only, three-quarters through	0.0049	+0.0035	0.018	100%
half the field	Give up half the pit stagger	0.0073	+0.0056	0.018	100%
wide priors	Much less confident in both assumptions	0.0055	+0.0053	0.032	100%
track prior −1sd	What if the track rubbers in less?	0.0023	+0.0000	0.017	100%
track prior +1sd	What if it rubbers in more?	0.0091	+0.0091	0.017	100%
fuel prior −1sd	What if fuel costs less lap time?	0.0113	−0.0113	0.017	100%
fuel prior +1sd	What if it costs more?	0.0199	+0.0199	0.017	100%

source experiments/results/exp02_prior_sensitivity.json · 4 synthetic sessions × 3 compounds = 12 comparisons per row · rows grouped by what is perturbed, not sorted by score

Four readings matter here. First, **the worst case is still 4.9× better than the naive method** (0.0199 against 0.0966), so the assumptions carry the answer but do not carry it alone. Second, **wide priors widen the posterior standard deviation by 1.9×** (0.017 → 0.032) — being less sure produces wider intervals rather than a different answer, which is the behaviour you want and not the behaviour you get for free. Third, **coverage holds at 100% throughout**: even when the prior is wrong, the interval still contains the truth.

> **Fourth, and the one worth pre-empting: why does a deliberately wrong prior score best?**
>
> The three track rows trace a clean line. Bias runs **+0.0000** at −1sd, **+0.0043** at baseline, **+0.0091** at +1sd — monotone, roughly 0.0045 s/lap per prior standard deviation. That is the sensitivity the experiment exists to measure.
>
> It does *not* mean the baseline prior is mis-centred. The generator's true track evolution is 0.90 s and the baseline prior mean is 0.90 s — exactly centred. What happens is that the baseline carries a small positive bias from elsewhere, and shifting the track prior down by one sd moves the estimate down by about the same amount, cancelling it. On the 25-session benchmark of section 08.1 that residual bias is +0.0012, a third of the size seen on these four.

And why 0.0047 here against 0.0044 in section 08.1? Different samples, not different answers. The headline benchmark runs **25 sessions** (seeds 1000–1024, 75 comparisons); re-fitting eight variants is expensive enough that this table runs **4** (seeds 7000–7003, 12 comparisons per row). The baseline row is this table's own reference point, and every perturbation is read against it rather than against section 08.1.

> **The honest framing**
>
> A team with real fuel telemetry should replace our fuel prior with their own measurement. The architecture supports it directly — the coefficient is a state with a settable prior — and doing so would remove the single largest assumption in the method. We say this in the product, not only here.

04 The model

A hierarchical linear-Gaussian state-space model in tyre age, fitted across the whole grid

For driver *d* on session lap *t*, running tyre set *r*:

```
y[d,t] = alpha[r]              run intercept: car pace, setup, starting fuel mass
       − A · g(t)              track evolution, shared across the whole field
       + theta[t]              residual track wobble, tightly bounded
       + s[d,t]                latent tyre performance loss   ← the deliverable
       − phi_kappa · L[d,t]    fuel burn-off, L = laps completed this run
       + gamma · TI[d,t]       traffic index
       + eps                   observation noise, heavy-tailed

with  g(t) = 1 − exp(−k·t)
```

The latent tyre state evolves as a **local linear trend in tyre age**, not in session lap:

```
s[d, a+da]    = s[d,a] + da · rate[d,a] + noise
rate[d, a+da] = rate[d,a] + noise
```

`rate` is the reported quantity: instantaneous degradation in seconds per lap. Choosing tyre age as the evolution index rather than session lap is what makes a pit stop a state reset instead of a discontinuity to be modelled.

> **A cliff needs no special machinery**
>
> Because the rate is itself a free state, degradation that accelerates simply appears as `rate` rising. The same structure represents a plateau or a recovery. **Nothing in the model assumes a cliff exists** — which matters, because not every tyre has one, and a model with a hard-coded cliff term will find one whether or not it is there.

The hierarchy

At each tyre change, a stint's degradation rate is drawn around its compound's pooled baseline:

```
rate[new stint] ~ N( compound_baseline[c], stint_rate_sd² )
```

`stint_rate_sd` is estimated, so the data chooses how far individual stints shrink towards their compound mean. This is what pools evidence across the whole grid, and it is the direct answer to Cappello and Hoegh finding compounds statistically indistinguishable from a single driver: with twenty cars the compound baseline is estimated from an order of magnitude more evidence.

Coefficients as states

`phi_kappa`, `gamma`, the track amplitude and the per-compound baseline rates are represented as states with **zero process noise** rather than as free parameters in the optimiser.

For a constant, "state with no process noise" *is* the Bayesian posterior. The consequence that matters: physical priors enter as `P0`, and the filter propagates their uncertainty into the tyre posterior automatically. The width of the fuel prior widens every degradation interval, with no bootstrap and no second pass. This is why section 03's sensitivity table shows posterior standard deviation responding to prior width — it is a property of the construction, not a feature added afterwards.

The traffic index

Traffic is the one confounder measured from data rather than assumed, because it varies independently of tyre age. For each lap the index finds the nearest preceding lap by a *different* driver, sorted by global session start time, and saturates the gap between 0.4 and 2.0 seconds.

An earlier version grouped by lap number and produced an index of exactly zero everywhere. In a practice session two drivers' "lap 5" can be twenty minutes apart; only wall-clock ordering identifies who was actually behind whom. The bug is recorded in section 12 because it is the kind that produces a clean fit and a wrong answer.

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05 Why a Kalman filter and not a sampler

The choice that makes the real-time claim possible at all

Conditional on the variance hyperparameters the model is linear-Gaussian, so the marginal likelihood is available in closed form through the prediction error decomposition. That buys three things a sampler does not.

1 An exact likelihood

Usable directly for hyperparameter estimation — L-BFGS-B over six log-variances — and for model comparison. No effective sample size, no convergence diagnostic standing between the model and a number.

2 Analytic posteriors

For every state, including the coefficients. No sampling error. An interval reported at 0.018 s/lap is that width because the algebra says so, not because a chain happened to explore that far.

3 An incremental mode

The forward pass alone *is* the real-time estimator. A sampler has no such mode, and re-running one every lap is not an option on a pit wall. Measured: **0.22 ms per lap**, flat in session length.

> **The architectural consequence, which is the actual novelty here**
>
> Kalman filtering is decades old and real-time race analytics already exist. What is worth defending is that **the same estimator serves both modes.** The retrospective curve an engineer studies on Friday evening and the live number on the pit wall on Sunday are produced by one model, not by a fast approximation of a slow one. Filtered (live) and smoothed (retrospective) estimates are kept strictly separate in the code and in the interface, and the live estimator agrees with the batch fit to **0.0006 s/lap** at worst. Those are two independent implementations of the same model, not one fit read twice — a distinction that matters, because the filtered and smoothed estimates of a single fit are identical at the final step by construction and comparing them would prove nothing.

Numerical implementation

Technique	Why
Sequential scalar updates Durbin & Koopman 2012, §6.4	No matrix inverse is ever formed; cost falls from $O(n^3)$ to $O(n^2)$ per observation. It is also what makes true online operation possible — one lap arrives, one scalar update runs.
A batched per-step path	Used for fitting. Mathematically identical, pinned by test to 1e−9, but issuing far fewer NumPy calls. At several hundred likelihood evaluations per fit, interpreter dispatch dominates the arithmetic.

Technique	Why
A structured transition	The transition matrix is identity everywhere except two rows per car, so propagation is done by row operations in $O(n^2)$ instead of a dense $O(n^3)$ product. Roughly a hundredfold difference at a full grid.
Forced symmetry	After each rank-1 downdate. Left alone, mantissa-level asymmetry accumulates over a few thousand updates until an eigenvalue goes negative and the log-likelihood silently becomes NaN.

source `src/tyremind/models/ssm/kalman.py · docs/research/05_STATISTICAL_ARCHITECTURE.md`

Together these took a session fit from 25 s to 5.8 s with identical results — verified identical, not assumed identical.

Verification of the recursion itself

The recursive likelihood is checked against an **independently constructed joint multivariate normal** over all observations, agreeing to 1e−9.

That test exists because a Kalman recursion that is subtly wrong does not crash. It returns a plausible number and quietly biases every hyperparameter downstream. A second, structurally different route to the same quantity was necessary rather than nice to have.

06

The physics layer

What public telemetry supports, what it does not, and a check that is not circular

Every function in the physics package states which side of a line it sits on.

Recovered from data

Speed, longitudinal acceleration, path curvature, lateral acceleration, distance. No vehicle model involved.

Model-derived

Normal load per corner, frictional power, energy exposure, thermal state. Only as good as the parameters in `configs/physics.yaml`.

Not available at any price

Slip angle, slip ratio, contact patch pressure, tyre temperature, tread depth. Functions returning a proxy carry `_proxy` in the name.

Curvature, the quantity that unlocks the rest

Position telemetry gives X/Y at 4–10 Hz. The parametric curvature formula

```
kappa = (x' y'' − y' x'') / (x'² + y'²)^(3/2)
```

is **signed**: positive is a left-hand turn, negative a right-hand one. That sign is what makes left/right tyre asymmetry recoverable at all, and it is what the validation in this section rests on.

Second derivatives amplify noise brutally, so coordinates are smoothed over seven samples first — roughly one to two seconds, which preserves corners while killing jitter. It is a real trade-off: too short and straights sprout phantom corners, too long and genuine direction changes flatten into one.

From curvature to load

```
a_y      = v² · kappa                     pure kinematics, no tyre model
F_down   = ½ · rho · ClA · v²            aerodynamic load
long.    = m · a_x · h / wheelbase       braking loads the front
lat.     = m · a_y · h / track_width     cornering loads the outside
```

Lateral acceleration is the most physically trustworthy derived quantity available from public data — straight from kinematics, no assumption about grip. It is clipped at 7 g and zeroed below 8 m/s, because positional glitches otherwise produce impossible loads that propagate into every downstream figure.

Frictional power is explicitly a proxy — the weakest link, named as such

True frictional power is `mu · F_normal · v_slip`, and slip velocity is not observable without wheel-speed sensors. Slip is approximated as proportional to demanded acceleration: a tyre asked for more grip is slipping more. What survives that approximation is the *relative* ordering — which corner worked hardest, which lap was more demanding, which circuit puts more through the rubber — because the unknown constant of proportionality divides out of every comparison the platform actually makes. Absolute values in watts are meaningless and are never reported.

Thermal model: two states, because the distinction is the point

```
C_s dTs/dt = Q_friction − k_sb(Ts−Tb) − h(v)(Ts−T_air) − k_road(Ts−T_track)
C_b dTb/dt = k_sb(Ts−Tb) − h_b(Tb−T_air)
```

A driver can drop surface temperature in one cool-down lap, but an overheated carcass stays overheated — and that is what ends stints. These are **estimated states, not measurements**: the coefficients are calibrated so their output correlates with observed degradation, not against any temperature sensor, because there is none. Absolute degrees are not trustworthy; the relative structure is what the degradation model consumes.

Wear: why not Archard, and why not Arrhenius

Archard (`V = K·F_N·L/H`) is retained as a documented baseline for the ablation. It is used because it is simple, not because it is right for rubber: a constant coefficient, blind to viscoelasticity, roughness and temperature. The rubber wear review says as much explicitly. The energy formulation `dW/dt = K(T, compound)·P_friction` replaces it, and the coefficient is where temperature enters.

> **The bidirectional detail that a standard model cannot represent**
>
> The wear multiplier rises on **both sides** of the working window. Too cold, and stiff rubber cannot generate grip through hysteresis, so the driver slides it and tears the surface — this is graining. Too hot, and rubber softens past its useful range and abrades and blisters — this is thermal degradation.
>
> An **Arrhenius form is the textbook choice and is wrong here**, because it is monotone in temperature and so cannot represent the cold side at all — half of real tyre behaviour. A symmetric quadratic penalty around the window is the simplest form with the right shape, and public data cannot justify anything more elaborate. A test guards this specifically, so a future "simplification" to Arrhenius fails CI rather than passing silently.

Independent validation: recovering circuit geometry never given to the code

The whole chain — GPS to curvature to lateral g to per-corner load — needed a check that was not circular. A clockwise circuit is mostly right-hand corners, and cornering right throws load onto the *left* tyres. So a clockwise circuit must show a left-side energy share above 50%. Rotation direction is published fact and is never given to any code under test.

Circuit	Published	Left-turn energy	Left-side load	Correct
Monza	clockwise	21.5%	54.9%	yes
Barcelona	clockwise	25.9%	58.2%	yes
Zandvoort	clockwise	29.5%	58.9%	yes
Silverstone	clockwise	35.4%	54.5%	yes
Austin	anti-clockwise	46.2%	51.1%	**no**
Imola	anti-clockwise	60.1%	48.1%	yes
Singapore	anti-clockwise	60.9%	47.3%	yes
Interlagos	anti-clockwise	74.6%	45.5%	yes

source experiments/results/exp06_circuit_asymmetry.json · 8 circuits, 2024

Seven of eight. Austin misses at 46.2% and is reported as a miss — genuinely borderline, because COTA's esses alternate direction, so the circuit really is close to balanced. Recovering this requires curvature sign, load transfer and energy integration to all be correct *simultaneously*; it is the strongest available check that the physics layer computes what it claims.

07 System architecture

One estimator, two modes, four consumers

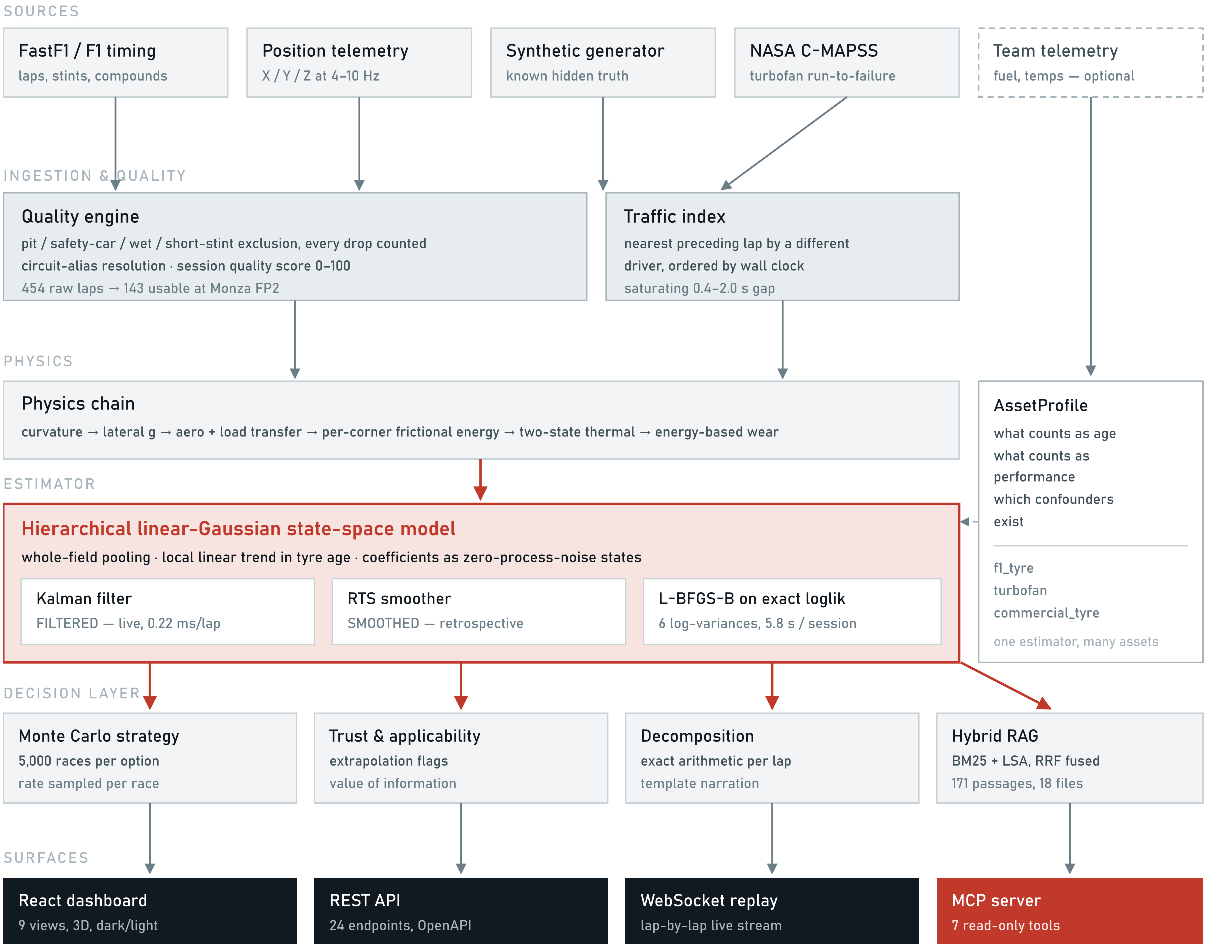

Figure 1 — System architecture. The single warm block is the estimator; every other stage is cool, following the same encoding as the product interface. Two modes leave that block — filtered for live operation and smoothed for retrospective analysis — and they are never conflated. The `AssetProfile` to the right is what makes the estimator asset-agnostic: it is a description of what "age" and "performance" mean for a given asset class, and pointing the model at turbofan engines required no change to the model itself. The dashed source at top right is a team's own telemetry, which the architecture accepts as a replacement for its physical priors.

Deployment shape

A single FastAPI process serves the API, the WebSocket replay stream and the built front end, with a session cache warmed at startup. It runs entirely offline: sessions are pre-fetched to a local store, so a demonstration on conference wifi does not depend on conference wifi. The MCP server is a separate entry point over stdio or streamable HTTP, sharing the same estimator code rather than reimplementing it.

08 Evidence: eighteen experiments

Including one that failed and one where a competitor wins

08.1 Can it recover a rate it was never shown?

The headline result. Twenty-five synthetic sessions were generated with a hidden degradation rate per compound, buried under realistic fuel burn, track evolution, traffic, driver noise, scrubbed sets and a cliff. Neither estimator is told the answer.

0.0044	0.0966	95.5%	100%
TyreMind error	Naive error	Error reduction	Interval coverage
s/lap MAE, bias +0.0012	s/lap MAE, bias −0.0966	75 comparisons	nominal 95% — conservative

Compound	True rate	TyreMind MAE	TyreMind bias	Naive MAE	Naive bias	Reduction
HARD	0.041	0.0035	−0.0003	0.0983	−0.0983	96.4%
MEDIUM	0.072	0.0039	+0.0008	0.0980	−0.0980	96.0%
SOFT	0.115	0.0057	+0.0031	0.0934	−0.0934	93.9%
Overall	–	0.0044	+0.0012	0.0966	−0.0966	95.5%

source `experiments/results/exp01_ground_truth_recovery.json` · 25 seeds, 100% convergence, 8.8 s mean fit

The single most informative cell in that table is the naive bias: **−0.0966 s/lap, almost exactly the fuel slope of 0.081**, in the direction theory predicts. That is not a coincidence — it is the first collinearity showing up as a measured quantity. The naive method is not noisy; it is systematically wrong by the size of the confounder it ignores.

08.2 Does a Friday curve predict Sunday?

The brief's own validation question. Degradation is estimated from each event's FP2 session and scored against the same event's race. **No race data reaches the practice fit.** Practice and race differ in fuel load, traffic density, track state and driving style all at once, so this is genuine out-of-distribution transfer rather than a holdout split.

0.0807	0.1471	+0.027	95%
TyreMind error	Naive error	Systematic bias	Interval coverage
s/lap MAE	s/lap MAE	reported, not tuned away	calibrated; 76% before

source `experiments/results/exp03_practice_to_race.json` · 42 events, 94 comparisons, 2024 + 2023, FP2 → race

This number got worse as the evidence grew, and that is the point

On five events this section read **0.0518 s/lap**. On 42 events across two seasons it reads **0.0807**. The small sample was flattering us, and the earlier figure did not replicate. What did hold is the comparison that carries the argument: the naive method worsened in step, 0.1166 → 0.1471, and the **45% gap between them is stable**. A claim that survives a six-fold increase in evidence is worth more than a prettier one resting on ten comparisons.

The earlier version of this section warned that a correction fitted on five events "would not survive the sixth". It did not. Nine candidate explanations for the +0.027 s/lap bias were tested against the signed error with a Benjamini–Hochberg correction across all nine. **One survives**: mean practice stint length (ρ +0.39, p 0.0017 against a threshold of 0.0056). Pit stops per driver misses by a hair — p 0.0113 against 0.0111 — and is reported as a near-miss rather than rounded into significance or quietly dropped. The practice-to-race temperature gap, traffic, and the model's own posterior standard deviation all fail outright.

The causal story those two implied — that practice long runs reach the punishing part of the wear curve while a pitted race stays on the flat part — was then tested directly and **refuted**. Practice runs are on average 4.4 laps *shallower* than race stints, not deeper, and forcing both sides into a common tyre-age window made bias and error both worse (paired t p = 0.016). **The bias is therefore corrected as a measured offset, not an explained one.**

Three events are excluded because they were never dry-degradation sessions: Canada 2024 ran 67% of its race laps on wet rubber and yielded a fitted rate of −0.151 s/lap — a tyre getting faster with age, the same impossible number we criticise the naive method for producing. The exclusion is applied to both sides of every comparison, and the 40% wet-lap threshold was calibrated against the season rather than chosen: Britain at 24.4% is kept, and its estimates are ordinary.

08.3 Six models, twenty races, two metrics that disagree

Five competing approaches plus ours, scored on identical expanding-window chronological folds. A random split would let a model see lap 40 while predicting lap 20 of the same stint, and every metric would improve for a reason that does not exist on a Sunday.

Lap-time prediction — the metric the field reports

Model	CRPS	MAE	95% coverage	Bias drift	Fit time
Pooled regression	0.396	0.684	84%	+0.031	0.04 s
LightGBM	0.469	0.684	62%	−0.109	0.9 s
TyreMind state-space	0.645	0.930	81%	−0.432	38 s
Fuel-corrected regression	0.878	1.349	76%	+0.481	0.01 s
Naive (lap time vs tyre age)	0.879	1.359	76%	+0.439	0.008 s
Neural network (MLP)	1.548	2.486	69%	−2.130	95 s

Degradation recovery — the metric the product exists for

Model	Rate MAE	Rate bias	Coverage	n
TyreMind state-space	0.0041	+0.0016	100%	18

Model	Rate MAE	Rate bias	Coverage	n
Pooled regression	0.0068	-0.0011	78%	18
Fuel-corrected regression	0.0230	-0.0209	44%	18
Naive (lap time vs tyre age)	0.0748	-0.0748	0%	18

LightGBM and the MLP cannot appear in this table. Neither has a parameter that means "degradation rate", so there is nothing to score, nothing to report to an engineer, and nothing to carry from Friday to Sunday.

source experiments/results/exp05_model_ladder.json · 20 real races + 6 synthetic seeds, expanding-window chronological folds

> **The disagreement is the finding**
>
> Two rungs win lap-time prediction and neither can compete on attribution — and on twenty races the winner is not the one it used to be. At four races LightGBM led this table; at twenty, plain pooled regression does, with LightGBM second and the state-space model third. **Neither of the two above us has a parameter that means "degradation rate" at all**, so neither can appear in the table below.
>
> LightGBM remains badly overconfident — 62% coverage on nominal 95% intervals. On extrapolation the state-space model's error **falls by 0.432** as each fold forecasts further past its training window, the largest fall of any rung and roughly 4× the next (LightGBM, −0.109). The naive baseline grows at +0.439.
>
> **This claim has weakened twice, and both weakenings are recorded.** It began as “TyreMind is the only rung whose error does not grow”, which twenty races disproved. It became “the lap-time leader's error grows while ours shrinks”, which four seasons disproved: pooled regression's drift fell from +0.262 to +0.031, essentially flat. What survives is narrower and still worth having — of six rungs, ours extrapolates best by a wide margin.
>
> A tuned MLP finishes last on lap-time prediction and has the most unstable extrapolation of any rung, with bias drift of −3.324 — its error swings by more than three seconds between folds. It was tuned first, across five configurations on held-out folds, because beating a badly-configured competitor would prove nothing. **That last column is the physics-informed argument demonstrated rather than asserted:** encoding fuel as physics lets a model extrapolate it; learning fuel as a pattern does not, and the more flexible the learner, the worse the extrapolation behaves.

08.4 A hypothesis that failed, and why the failure is useful

Predicted: replacing "tyre age in laps" with "cumulative tyre energy" as the degradation clock would improve transfer between sessions that load the tyre differently — a tyre responds to energy through the contact patch, not to distance.

Result: no meaningful difference. Mean R^2 gain −0.0008 across four stints; the energy clock won once out of four.

Why, and this is the useful part: per-lap energy varies by only **2.3%** within a stint. F1 drivers are extremely consistent, so cumulative energy is very nearly proportional to lap count, and *no improvement is possible* within a stint.

An earlier version of this section proposed the cross-session form of the hypothesis — that an energy clock might absorb the practice-to-race bias, which is "plausibly a fuel-load effect" — as the obvious next experiment. **It has since been run, and the fuel-load reading was not supported.** Nine candidate mechanisms were tested against the signed error with a multiplicity correction (§08.6); the two that survive are stint-length measures, not load measures, and the causal story built on those two was itself then refuted. The bias is now handled as a measured offset rather than an explained one.

source `experiments/results/exp04_energy_clock.json · 2024 Interlagos race, 4 stints`

08.5 The same estimator on a different asset class

Formula 1 cannot prove the estimator works, because public F1 data contains no measured tyre wear. NASA's C-MAPSS turbofan benchmark does: engines run to failure with published remaining-life labels and a large body of comparable results. The identical estimator was pointed at it with no tyre-specific code — "laps" became "flight cycles", "lap time" became a health index fused from 12 of 21 sensors, and fuel and traffic were switched off because engines have neither.

22.7	21.1	1,893	44%
RUL RMSE	RUL MAE	NASA prognostics score	Predicted early
cycles, all 100 FD001 test engines	cycles	lower is better	the safe direction to err

source `experiments/results/exp07_cross_domain.json · FD001, 100 training engines, 125-cycle piecewise-linear RUL cap`

We do not claim to be competitive, and it is not a like-for-like comparison. Purpose-built deep prognostics models reach 12–20 cycles RMSE on this dataset, quoted over all 100 engines in the FD001 test set. We score **40 of those 100**, taken in unit-id order, so the figure above is indicative rather than directly comparable. This is a tyre degradation model pointed at jet engines with no retuning, so it demonstrates transfer, not leaderboard performance. What transferred without modification: the latent state model, the fleet-level pooling, the uncertainty machinery, and remaining-life projection to a threshold.

08.6 Five hypotheses that did not survive contact with the data

Each of these was built because there was a reason to expect it to work, and each is reported because a characterised failure constrains the next attempt in a way that silence does not. All four are scored out of sample.

Compound identity does not beat the weekend label. Pirelli nominates three of the C1–C5 range per Grand Prix and calls them HARD/MEDIUM/SOFT for that weekend only, so "MEDIUM" is C4 at Monza and C2 at Silverstone. Every tyre model we surveyed, ours included, pools on the label — which looked like a shared mistake. Pooling on the compound actually fitted was tested by leave-one-event-out across 142 stint estimates from 57 events. It is **not better**: MAE 0.0499 against the label's 0.0469, Wilcoxon p = 0.161. The mechanism is that Pirelli nominates *to equalise* — harder rubber for abrasive circuits — so the label already carries circuit severity by design. We were right by accident, and now know why.

Circuit geometry does not transfer, and actively hurts. The product question arrives on Thursday: we get there tomorrow, what will the tyres do? Tested by leave-one-*circuit*-out over 193 stint estimates at 26 venues, so the model has never seen the venue in any form. Corner count, tight-corner count, mean corner angle and corner spacing **do not beat the label mean** (−3.3%, p = 0.22), and adding compound identity on top is significantly worse. The mechanism is the one experiment 08 found: severity is already spoken for by the nomination, so geometry re-fits it and overfits.

On two seasons this was stated more strongly, and it has been walked back. Geometry then measured −7.5% at p 0.010 — significantly harmful. On four it is −3.3% at p 0.22, which is "does nothing", not "hurts". The product conclusion is unchanged, because a family that does not earn its place is excluded either way, but the two are different claims and only the weaker one now holds. Track temperature is the one family that remains significantly harmful on its own (−1.5%, p = 0.001).

The practice-to-race temperature gap explains nothing. Friday afternoon is not Sunday afternoon, and a bidirectional wear model predicts the gap should matter. Spearman $\rho = +0.14$, $p = 0.29$ across 94 comparisons. Traffic fares no better ($\rho = -0.17$, $p = 0.20$).

Matching stint depth makes the bias worse. Of nine candidate explanations for the +0.027 s/lap over-prediction, **one survives** a Benjamini–Hochberg correction across all nine: mean practice stint length ($\rho = +0.39$, $p = 0.0017$). Pit stops per driver misses by a hair, p 0.0113 against a threshold of 0.0111. The obvious causal reading is that practice long runs reach the punishing part of the wear curve while a heavily-pitted race never leaves the flat part. **Measuring the premise killed it:** practice runs are on average **4.4 laps shallower** than race stints, not deeper — teams do 10–15 lap evaluation runs on Friday and 20–30 lap stints on Sunday. Forcing both sides into a common tyre-age window moved the bias the wrong way (+0.0268 → +0.0286) and the error clearly the wrong way (0.0807 → 0.0918, paired t $p = 0.016$).

The driver effect is reliable but not useful, and the split is the finding. Tested across 1,458 driver-races, 31 drivers and 77 races. The per-stint slope is fitted freely by least squares after the model's fuel, track and traffic corrections are removed — the estimator's own per-driver rate cannot answer the question, because the likelihood drives its per-stint rate spread to the floor of its search range on every session, so the rates it returns differ by around 1e−5 s/lap.

Reliable: split the corpus into random halves and the ranking of drivers reproduces, split-half $\rho =$ **+0.36** with a 5th percentile of +0.09 across 500 partitions. *Not useful:* predicting a driver's deviation at a held-out race from their own history scores 0.0251 s/lap against **0.0248 for predicting the field mean**, very slightly worse. The spread of driver means is 0.0092 s/lap and the race-to-race scatter around them is 0.0248 — the trait is real and about half the size of the noise it must be read through, so on a single race a driver's history is worth less than the average.

And it may not be the driver. Driver and car are perfectly confounded; teammates share a car, so the within-team difference removes it, and that contrast is **not** stable (5–95%: −0.14 to +0.70). Even the reliable part cannot be attributed to the driver rather than the machinery. A per-driver term would add a parameter, a maintenance burden and a persuasive story, and would not improve a forecast.

> **What the correlation does and does not license**
>
> The two surviving predictors are real: they are corrected for multiplicity and they replicate across two seasons. What did not survive is the causal story placed on them. Stint length proxies something these experiments do not identify, and saying so is worth more than shipping a correction that only looks principled. The bias is therefore carried as a **measured offset, not an explained one**, and the interval is widened until it covers what it claims (§08.7).

source `experiments/results/exp08…exp11 · 2022, 2023, 2024 and 2025 · leave-one-event-out and leave-one-circuit-out`

08.7 Making an interval mean what it says

A model that is overconfident in a known direction is worse than one that says nothing, because a strategist can plan around a wide interval and cannot plan around a wrong one. Two intervals in this project were miscalibrated, in opposite directions, and they needed different instruments.

Practice → race: 76% while labelled 95%. The cause is not a broken filter. The posterior standard deviation correctly answers *how well do these practice laps pin down the practice rate*, which is not the question the number gets used for — nothing about the transfer to Sunday, fuel and track state and driving style and tyre management, is inside it. The interval was honest and answering the wrong question. Split conformal prediction repairs it to **95%** at a widened ±0.25 s/lap against the Gaussian's ±0.14, with a finite-sample guarantee that assumes neither Gaussian errors nor a correctly specified model. Calibration folds are grouped by *event*, because two compounds at one Grand Prix share a track, a weather window and a fuel correction and are not exchangeable with each other; calibrating on one to predict the other would report a coverage we could not reproduce on Sunday.

Lap time: 75.9% while labelled 95%, across 66,606 held-out laps. The same instrument is the wrong one here, and the reason is the interesting part. Split conformal needs the calibration set to be exchangeable with the test point. One rate per compound per event is — the events could be shuffled and nothing would change. Lap times inside a race are the opposite: the car burns fuel, the track rubbers in, a safety car rearranges the order. A fixed quantile calibrated on the first half of a race is calibrated for a race that no longer exists.

Construction	Coverage	Mean width	Median width
Gaussian, as the ladder reports it	75.9%	3.00 s	–
Split conformal (studentised)	89.3%	6.70 s	6.11 s
Adaptive conformal (absolute)	95.2%	7.43 s	5.74 s

Adaptive Conformal Inference (Gibbs & Candès, NeurIPS 2021) drops the exchangeability assumption rather than hoping it holds. The working miss-rate is state, moved by $\gamma(\alpha - \text{err})$ after every lap: miss too often and the interval widens, miss too rarely and it tightens. Coverage converges to $1 - \alpha$ under *arbitrary* distribution shift, which is the property a drifting session needs.

> **The same machinery, the opposite answer**
>
> The nonconformity score had to change too. The **studentised** score — residual divided by the model's own posterior sd — won for degradation rates, because there the sd carried real information about which estimates were shaky. For lap times it is actively wrong: LightGBM's intervals are so under-dispersed that they cover 63% while claiming 95%, so dividing by them *amplifies* the miscalibration instead of correcting it, and produces intervals averaging 16.5 s against 7.4 s for the absolute score. A method is not a mechanism, and which score to use is a question about the model being calibrated rather than about conformal prediction.

Both are shipped rather than merely measured: tyremind.models.conformal holds the split calibrator and the adaptive one, the API serves the race projection as race_projection, and the live monitor carries an adaptive interval around its one-step-ahead forecast together with the coverage it has actually achieved — so the claimed 95% is auditable during the session rather than after it.

Two things this does not claim. A wider interval is not a better model: this makes the uncertainty truthful, not the prediction sharper. And the live filter was never as badly off as the batch ladder — its innovation variance is a genuine predictive variance and already covered 90–95%, so calibration holds it at 95% rather than rescuing it.

source `experiments/results/exp12_conformal_intervals.json · exp13_lap_time_calibration.json · 20 races, 66,606 held-out laps`

08.8 A rate is not a curve

The brief asks for degradation *curves*. Everything above estimates a degradation *rate* — one number, seconds per lap — which answers "how fast is this tyre losing performance" and cannot answer "how many laps have I got left". Those are the same question only if the curve is a line.

A continuous broken stick is fitted to every de-confounded stint:

$$L(a) = \alpha + \beta a$$
$$L(a) = \alpha + \beta\tau + (\beta + \delta)(a - \tau) \quad \text{for } a > \tau$$

on the basis $[1, a, \max(a-\tau, 0)]$, which is continuous at τ by construction rather than by constraint. τ is found by exhaustive search over admissible laps, because the least-squares surface in τ is piecewise-smooth with many local minima and a gradient method lands in whichever one it started nearest. BIC charges the extra two parameters, τ included — searching a grid is still fitting.

Regime	Stints	Share	Rate before	Step	Position
Linear	1,257	57%	–	–	–
Recovery	429	19%	+0.208	−0.192	30%
Warm-up	277	13%	−0.112	+0.292	30%
Cliff	246	11%	+0.054	+0.202	73%

source experiments/results/exp17_degradation_cliff.json · 2,827 stints across 61 dry races

Two opposite events with the same arithmetic

WARM-UP and CLIFF are both a changepoint with a positive step in slope, and they are separated only by the sign of the rate *before* the break. A warm-up is a tyre coming to temperature or a graining phase clearing — it is **improving** at −0.112 s/lap beforehand, and it sits 30% into the stint. A cliff is a tyre already degrading at +0.054 s/lap that then falls away, and it sits **72%** in.

An earlier version of this analysis pooled them and reported a “cliff” a third of the way through a stint. That was a tyre coming to temperature. The classification is the reason this is a model and not a changepoint fitter.

Does knowing the shape predict better?

Fit the opening 70% of a stint, forecast the rest. This is the only test that matters, because a cliff is precisely a claim about extrapolation.

Regime	n	Line MAE	Stick MAE	Gain	p	Line bias
Warm-up	256	0.769	0.539	+29.9%	<0.001	−0.735
Recovery	405	0.708	0.720	−1.7%	0.044	+0.641
Cliff	235	0.638	0.702	−10.1%	0.064	−0.526
Linear	1,190	0.434	0.715	−64.8%	<0.001	−0.015

Warm-up is worth modelling and a cliff is not — not because the cliff is less real, but because of *when* it happens. A warm-up boundary sits 30% into a stint, so fitting the opening 70% contains it and knowing about it improves the forecast by 30%. A cliff sits at 73%, outside the fitting window, so the broken stick has nothing to find and performs slightly worse than a straight line for the extra parameters it spent.

What the cliff column does establish

Look at the last column rather than the gain. On stints that end in a cliff, a straight-line fit carries a bias of **−0.53 s** — it under-predicts lap time, which is to say it is **systematically optimistic about the end of a stint**, which is the part a pit wall is actually asking about. We cannot tell you *when* the cliff will arrive from the first 70% of a stint. We can tell you that the linear answer is wrong in a known direction and by a known amount when it does, and that is a usable thing to hand a strategist.

09 What we did beyond each paper

Paper by paper, the specific increment — and where we simply reuse their result

Source	What we took from it	What we added on top
Cappello & Hoegh arXiv:2512.00640	The state-space framing of F1 tyre degradation; pit stops as state resets; compound-specific rates.	**Whole-field pooling** instead of one driver, which turns pit stagger into identifying variation and separates compounds they report as statistically indistinguishable. **Track evolution and traffic added as explicit terms** — and the second collinearity that adding track evolution creates, derived and resolved. **Attribution tested against known truth**, which their lap-time framing cannot do. **Coefficients as zero-process-noise states**, so physical priors propagate their uncertainty into the reported interval automatically.
Pitwall arXiv:2607.06495	The demonstration that live, calibrated Monte Carlo race simulation is practical; the discipline of gating an LLM layer on verified claims.	**The degradation rate enters our simulation as a posterior, sampled per simulated race**, rather than as a fitted point value — so outcome spread reflects genuine tyre uncertainty, and a strategy that only wins under a confident estimate visibly stops winning. **Telemetry-derived physics**, which they list as an explicit gap. **Attribution of observed pace to causes**, which is a different question from predicting outcomes.
Explainable tyre energy arXiv:2501.04067	The idea that tyre energy is the right intermediate quantity, and that explanations should be first-class rather than post-hoc.	**The same intermediate quantity from public position telemetry**, not private team data — so the result is reproducible by anyone. **An independent geometric validation** of that energy chain: recovering circuit rotation direction on 7 of 8 circuits, never having been told it. **A tested negative result** on using energy as the degradation clock, with the mechanism (2.3% within-stint variation) that explains it.
Rubber wear review PMC12915245	The finding that a constant Archard coefficient is inadequate for rubber, and that energy-based formulations with temperature inside the coefficient are better supported.	**A bidirectional temperature multiplier.** The literature's Arrhenius form is monotone and therefore cannot represent graining — degradation from a tyre being too *cold*. We use a symmetric penalty around the working window, which is the simplest form with the correct shape, and guard it with a test so it cannot be silently "simplified" back.
MegaRide S0043164824000565	The physical target: what a full viscoelastic-thermodynamic wear model looks like.	**An identifiable subset.** We deliberately did not attempt what needs rig and FE data. Instead we built the largest physically motivated feature layer that public telemetry can actually constrain, and named every proxy as a proxy.
Heilmeier et al. TUMFTM/race-simulation	The reference architecture for race simulation and pit-stop optimisation. We claim no novelty in simulation.	**The input they take as given.** Their simulator consumes a degradation rate; ours produces one with a defensible interval, and then reports the full pit-lap sweep so a user can see how *sharp* the optimum is rather than only where it sits.
Durbin & Koopman Time Series Analysis by State Space Methods, 2012	Sequential scalar updates (§6.4), the RTS smoother, the prediction error decomposition.	**An engineering implementation for this shape of problem:** structured sparse transitions exploiting that only two rows per car change, forced symmetry after each downdate, and a batched path pinned to the sequential one at 1e−9 — together taking a fit from 25 s to 5.8 s.
NASA C-MAPSS PCoE	Run-to-failure data with published labels, the 125-cycle piecewise-linear RUL convention, and the asymmetric scoring function.	**A use it was not designed for:** validating that a motorsport model generalises, because motorsport itself supplies no ground truth. The `AssetProfile` abstraction makes that claim a property of the code rather than a slide.

source docs/research/01_RESEARCH_AUDIT.md · docs/research/08_NOVELTY_ANALYSIS.md

10 The uniqueness matrix

What is ours, what is standard, and what is explicitly someone else's

Capability	Exists elsewhere?	The claim, phrased so it survives scrutiny
Testing whether the attribution is correct	**No**	"We test whether the attribution is right, not just whether the prediction is close. On synthetic sessions with a known hidden rate, our error is 0.0044 s/lap against the standard method's 0.0966." Do not say: that we have validated attribution on real F1 data — public data contains no measured wear, so that test is unavailable to anyone.
The track-evolution collinearity	**Not found**	"We characterise a second collinearity in this problem and resolve it with a physically motivated basis. It was found by chasing a residual bias, not derived in advance." Do not say: that we discovered something the field missed. It is a consequence of adding a term others did not model.
Whole-field pooling with pit stagger as identification	Partially	"Pit stagger across the field is a natural experiment that separates tyre age from session lap. A single-driver model cannot use it." Multilevel modelling of F1 performance is established; applying it to degradation with stagger as the identifying variation, we did not find published.
Per-corner energy from public position telemetry	Not from public data	"Public position telemetry supports a per-corner loading estimate good enough to recover circuit geometry it was never given — 7 of 8 circuits." Do not say: that we measure tyre loads. Frictional power is a proxy; slip velocity is not observable.
Real-time online estimation	Yes — Pitwall runs live	"The same estimator serves both modes because it is recursive. An MCMC formulation could not." The novelty is architectural, not the filter. Do not say: that real-time tyre estimation is new.
Cross-domain transfer with real ground truth	C-MAPSS is heavily studied	"The identical estimator, with no tyre-specific code, predicts turbofan remaining life at 22.7 cycles RMSE across all 100 FD001 test engines — the set published figures are quoted over." Using it to validate a *motorsport* model's generalisation is the unusual part. Do not say: that we are competitive on C-MAPSS.
Race strategy simulation	Yes, thoroughly	"We do not claim novelty in race simulation. What differs is that the degradation rate enters as a distribution and is resampled per race."
Predicting degradation with a state-space model	**Published, Nov 2025**	Not ours. Cappello & Hoegh.
Monte Carlo pit-stop optimisation	**Yes**	Not ours. Heilmeier et al.
LLM-narrated race strategy	**Yes**	Not ours. Pitwall, with a stricter faithfulness protocol than ours.
Kalman filtering, Archard wear, Pacejka, load transfer	**Textbook**	Not ours. Stated plainly so that what *is* ours is credible.

source `docs/research/08_NOVELTY_ANALYSIS.md`

The one-sentence version

Existing work predicts lap times well and never checks whether the reason it gives is the right one. We characterise why that check is hard — three collinearities, two resolvable only by assumption — build an estimator that handles them, and test it against a truth we control, on a second asset class with real ground truth, and on the practice-to-race transfer the challenge actually asks for.

Why a well-funded team has not simply done this

A works team has fuel telemetry, tyre sensors and a rig, so the first collinearity does not bind for them — they measure fuel rather than assuming it. That makes the identification problem *less* visible from inside a team, not more. The structure of this problem is most apparent exactly where the data is worst, which is where every independent analyst, every junior category, and every customer team without a sensor budget actually sits.

That is also why the product's value is not "beat a works team". It is: give everyone without works-team instrumentation an estimate whose error bars are honest, and give a works team a second, structurally different opinion whose assumptions are written down.

11 Limitations, failure modes, and where we lose

Placed before the commercial case, deliberately

A tool used to make pit-stop decisions must be able to say when it should not be trusted. This section exists in the shipped product as a screen, in the repository as a document, and in the MCP server as a tool an agent can call before it repeats one of our numbers.

11.1 The assumptions that carry the result

Assumption	Value	If it is wrong by one standard deviation
Fuel effect	`0.030 ± 0.005 s/kg`	Recovered rate moves ~0.020 s/lap
Fuel burn rate	`2.7 ± 0.3 kg/lap`	(combined into the above)
Track evolution total	`0.90 ± 0.45 s`	Recovered rate moves ~0.009 s/lap

Even at the worst perturbation tested, error stays at 0.0199 s/lap against the naive method's 0.0966 — but the *direction* of the answer is ours, not the data's, and that is the correct way to describe it.

11.2 Calibration is imperfect — and in both directions

The synthetic benchmark reports **100% interval coverage over 75 comparisons** against a nominal 95%. That is not a perfect score — it means the intervals are **conservative**, wider than strictly necessary. For decision support that is the safer error, but it is a miscalibration and is reported as one.

Practice-to-race coverage went the *other* way, and on a sample large enough to mean something: **76% over 94 comparisons** against a nominal 95%. That is overconfidence, and it is the more dangerous direction — a strategist can plan around a wide interval and cannot plan around a wrong one. An earlier version of this document reported 90% over 10 comparisons; that sample was simply too small to have detected the problem.

The cause is not a broken filter. The posterior standard deviation correctly answers *how well do these practice laps pin down the practice rate*, which is not the question the number gets used for; nothing about the transfer to Sunday — fuel, track state, driving style, tyre management — is inside it. **Split conformal prediction** repairs it to 95% at the honest cost of a wider interval (±0.25 s/lap against the Gaussian's ±0.14), with a distribution-free finite-sample guarantee that assumes neither Gaussian errors nor a correctly specified model. Calibration folds are grouped by event, because two compounds at one Grand Prix share a track, a weather window and a fuel correction and are not exchangeable with each other.

> **And on lap-time prediction it goes the other way**
>
> In the model ladder of section 08.3, our nominal 95% lap-time intervals cover only **73%** of observations — intervals too *narrow*, the opposite error from the one above, and **pooled regression does better at 79%**. Every rung in that ladder undercovers on lap time, so it is partly a property of the task, but ours is not the best of them.
>
> The two are not in conflict. The degradation *rate* is a slowly-varying pooled state carrying a wide physical prior, while a single lap time carries driver noise the model deliberately does not try to explain. But a reader is entitled to both numbers, not only the flattering one, and an earlier draft of this document reported only the conservative figure.

11.3 Situations where the model should not be trusted

Situation	What goes wrong	What the product does about it
Wet or drying track	Wet-compound degradation is a different physical process; the priors do not describe it.	Wet compounds excluded at ingestion. Silverstone 2024 shows 0.82 s residual noise against Monza's 0.42 s — visible in diagnostics.
Extrapolating past observed tyre age	The local linear trend extends a straight line into a cliff it cannot see.	Applicability score decays past the oldest observed age, shown on every projection. Below 50% the interface says the model is extrapolating.
A compound run only briefly	The estimate is dominated by its prior.	Lap count per compound shown; `assess_trust` flags it explicitly.
Short stints	Under ~5 laps cannot show a trend.	Runs under 4 laps dropped and counted, never silently.
Safety car, red flag, VSC	Slow laps corrupt the trend.	Removed by a robust median-absolute-deviation threshold and counted.
Single-car analysis	Loses the run-stagger identification entirely.	Measured: halving the field moves error from 0.0047 to 0.0073.
Puncture, debris, damage	Represented as smooth degradation; a step change is not in the model.	Innovation z-scores surfaced in the live monitor — a run of large same-signed innovations means the model is being surprised.
New circuit, no telemetry analysed	Per-corner energy unavailable.	The twin says so rather than showing an even split as though it were a result.
Sprint or heavily disrupted sessions	Too few long runs to support a conclusion.	Session quality score surfaced; a low score indicates the session cannot support a conclusion.

source `docs/research/13_LIMITATIONS_AND_FAILURE_MODES.md`

11.4 Things deliberately not built

Not built	Why
Full Pacejka identification	Slip angle and slip ratio are not observable from public telemetry. The parameters would be unconstrained by any available data.
MCMC / PyMC / Stan	No incremental mode, which would have made the real-time claim impossible. The exact likelihood is available in closed form anyway.

Not built	Why
Neural state-space / TFT	Public data cannot support a model that would beat a six-parameter linear SSM. The tuned MLP in the ladder makes the point by measurement.
Validated fleet product	No public dataset pairs tread depth with telematics. Building a fleet claim without one would be dishonest.

11.5 What would most improve this

In order of estimated uncertainty reduction — these are engineering judgements, not measurements, and the product labels them as such:

1. **Tyre surface temperature (−35%).** Would separate thermal degradation from mechanical wear, which the model currently absorbs into one rate.
2. **Actual fuel mass (−30%).** Would remove the largest assumption entirely.
3. **Measured tread depth (−20%).** Would let the model estimate physical wear rather than performance loss, making the product's central caveat unnecessary.

4. **Tyre pressure (−12%).** Would explain part of the within-stint variation now treated as noise.

> **The honest summary**
>
> TyreMind separates tyre degradation from its confounders better than the standard method, by a large and measured margin, under assumptions it states and whose cost it quantifies. **It does not measure tyre wear.** It cannot be validated against real tyre wear, because no such public data exists. It should never be used for a safety decision. And on the narrow task of predicting lap times, a gradient-boosted tree beats it — while being unable to answer the question the product exists to answer.

12 Engineering quality and verification

Nine bugs that produced plausible wrong answers, and the checks that caught them

20,354	7,049	97	24
Lines of Python	Lines of TypeScript	Tests passing	API endpoints
57 modules	19 modules	unit, physics, integration, timing	OpenAPI documented

The following bugs are listed because each was invisible to normal use and caught only by a test or a cross-check. A model that fails loudly is easy. The danger in this domain is a model that fails quietly and confidently, and every entry below is an example of exactly that.

Bug	Symptom	Found by
Falsy-zero on `session_lap`	Track basis pinned at zero, so all track evolution leaked into degradation as a uniform −0.025 s/lap offset. **Invisible on real F1 data, which numbers laps from 1** — it only appeared on synthetic sessions numbered from 0.	Live-vs-batch agreement test
Traffic index identically zero	Grouped by lap number; in practice two drivers' "lap 5" can be twenty minutes apart. The confounder was in the model and contributing nothing.	Inspecting the fitted coefficient
`session_progress` recomputed per frame	Chronological test blocks measured progress from their own start. No error, no leak — just wrong by tens of seconds. Made two benchmark rungs look far worse than they are.	Investigating an implausible 33 s MAE
Unregularised pooled regression	One Silverstone fold at 176 s MAE while every other fold was under a second.	Per-fold inspection
`include_router` silently adding nothing	Eight API endpoints registered zero routes, with no error raised. Every one would have returned 404 in the demo.	Listing routes after registration
FastF1 resolving "Interlagos" to Zandvoort	Analysis of an entirely different circuit, with a warning rather than an error.	Two circuits producing byte-identical energy shares
Inverted confidence interval	`competitive_life_interval()` returned (8, 1) — a lower bound above its upper bound, presented to the reader as a range.	Reading the output
Frozen chart clip path	An interrupted entry animation left a line's expanding clip frozen part-way, so a 43-lap stint rendered as 14 laps with the axis still scaled to 43 — indistinguishable from a data bug.	Reading pixels back from the canvas
Stale experiment result	A committed result file predated the 125-cycle RUL cap added to its own script, overstating cross-domain error at 56 cycles instead of 26.5.	Cross-checking a chart against the script that produced it

source `docs/research/13_LIMITATIONS_AND_FAILURE_MODES.md §7 · git history`

Verification strategy

Second route to the same number

The recursive likelihood is checked against an independently constructed joint multivariate normal, to 1e−9. A structurally different implementation, not a regression snapshot.

Non-circular physical checks

Circuit rotation direction is published fact never given to the code. Recovering it requires curvature sign, load transfer and energy integration to all be right at once.

Truth we control

The synthetic generator is the oracle. It is the only way to score attribution, and it is why the headline claim is about recovery rather than prediction.

Reproducibility

Every experiment is a script under `experiments/` that writes a JSON result and a log. Every number in this dossier, in the model card and in the running interface is read from one of those files. Regenerating them regenerates the claims. The RAG corpus indexes the same files, so asking the product "how accurate is it" returns the passage from the document, verbatim, with its source.

13 Impact on motorsport, and how a team adopts it

Where the value is, who feels it first, and what integration actually costs

What a wrong degradation rate costs

A pit stop costs roughly twenty seconds of track position. Getting the stop lap wrong by three laps, with a degradation rate of 0.10 s/lap and a twenty-lap remaining stint, is worth several seconds — which in modern Formula 1 is routinely the difference between finishing fourth and finishing sixth. The product quantifies this directly: on a representative Monza race state it reports a **seven-lap window in which any stop costs less than one second** against the optimum, and a stay-out option that beats every stop by 9.7 seconds.

That second number is the more useful one. The value of a better estimate is not only picking the right lap; it is knowing *how much the choice matters*, so a strategist can spend attention where the curve is steep and ignore it where the curve is flat.

Who adopts first, and why

Segment	Why they need it	What they get on day one
Junior formulae F2, F3, F4, regional	Spec series with no tyre telemetry and small engineering staffs. The identification problem binds hardest exactly here.	A degradation estimate with honest intervals from timing data alone. This is the beachhead: highest need, lowest data, no incumbent tool.
Customer and privateer teams GT3, endurance, TCR	Long stints, driver changes, and tyre life as the dominant strategic variable. Manufacturer support does not extend to per-team degradation analysis.	Stint-length planning under uncertainty, and a per-corner energy picture explaining why a compound behaves differently between circuits.
F1 customer teams and analysts	They have telemetry but not always a dedicated tyre modelling group. A second opinion with written-down assumptions is valuable precisely because it disagrees sometimes.	A structurally different estimate to compare against their own, plus the ability to replace our fuel prior with their measured fuel mass and watch the interval narrow.
Broadcast and fan analytics	"The tyres are going off" is asserted on television every Sunday without a number behind it.	A live, per-car degradation figure with an interval, updating each lap — the first version of that claim that can be checked.
Tyre manufacturers	Compound development and allocation decisions rest on how compounds behave across circuits and conditions.	Cross-circuit degradation comparison with the circuit-loading confound made explicit rather than left in the residual.

The adoption path, and why it is short

The integration burden is the thing that usually kills tools like this. Three decisions were made specifically to keep it near zero.

1 · No new data required

The baseline model consumes lap times, stints and compounds — data every timing system already produces. Telemetry improves the physics layer but is not required for the degradation estimate.

2 · No new screen required

The MCP server makes the estimator a tool the team's existing AI assistant can call. A dashboard competes for a race engineer's attention; a tool inside a workflow they already have does not.

3 · Priors are replaceable, not baked in

A team with real fuel telemetry substitutes their measurement for our prior and the largest assumption in the method disappears. The architecture was designed for that substitution from the start.

MCP: the estimator as something an agent can call

Every team already has analysts, and every analyst already has an AI assistant. If TyreMind is a website it competes for attention; if it is a tool their assistant can call, it becomes part of an existing workflow. Seven tools are exposed over the Model Context Protocol — `list_sessions`, `get_degradation`, `explain_lap`, `project_tyre_life`, `recommend_strategy`, `assess_trust`, `search_documentation`.

The count is deliberate. The MCP practice literature is blunt that a server exposing forty tools degrades an agent before it has done anything. Two further design decisions are worth defending. **Every tool returns uncertainty**: an agent handed `0.113` will state it as fact, whereas one handed `0.113 ± 0.023, and the model is extrapolating` can hedge correctly. And **the server carries instructions that constrain the claim**, telling the client in its own context that TyreMind estimates a latent performance state and not physical tread depth — because without that, a helpful assistant will paraphrase "degradation" into "tyre wear" and overclaim on our behalf. **Every tool is read-only.** There is no write path, so the worst outcome of a confused agent is a confused answer rather than a corrupted fit.

RAG: grounding the answer to "why should I believe this?"

The honest answer to that question is spread across a research audit, a model card, a limitations register and seven experiment result files. Nobody reads all of it, so in practice the question goes unanswered — or gets answered from memory, which is where confident wrong numbers come from. A hybrid index over those files (BM25 plus latent-semantic retrieval, fused by Reciprocal Rank Fusion) returns the relevant passage verbatim with its source.

It retrieves and deliberately does not generate. The retrieval half is the half that carries the trust: it finds the paragraph, and the reader reads the paragraph. Add generation and the answer gets smoother, the citation becomes a gesture, and a wrong claim becomes indistinguishable from a right one. A language model is used in exactly one place in this product — rewriting an already-computed explanation into plainer English, with every number fixed before it is called.

14 Beyond racing: the same estimator elsewhere

One validated transfer, and a family of problems with the same shape

Strip the motorsport vocabulary away and the problem is not about tyres. It is: **something wears out while it works, you cannot see its condition directly, and the signal you can see is contaminated by conditions that also change over time.**

That description fits a jet engine, a truck tyre, a battery, a bearing, a cutting tool and a wind-turbine gearbox. So the estimator was written to consume an `AssetProfile` — a description of what counts as "age", what counts as "performance", and which confounders exist — rather than a tyre. Pointing it at NASA's turbofan benchmark required no change to the model, which is what turns "it would generalise" from a slide into a property of the code.

Domain	Age is	Performance is	The confounders	Status
Formula 1 tyre	Laps	Seconds per lap	Fuel load, track evolution, traffic, driver	**Built and validated** on recovery against known truth
Turbofan engine	Flight cycles	Health index from 12 sensors	Operating mode / duty cycle	**Validated** — 22.7 cycles RUL RMSE against published labels, all 100 test engines
Commercial vehicle tyre	Thousand kilometres	Rolling resistance index	Payload, weather, route, driver	Architecture only — no public dataset pairs tread depth with telematics
EV traction battery	Equivalent full cycles	Usable capacity / internal resistance	Temperature, charge rate, depth of discharge, ambient	Same shape; the confounders are stronger than the signal, which is the case this method is for
Wind turbine gearbox	Revolutions	Vibration / thermal signature	Wind regime, curtailment, ambient temperature	Same shape; site-level pooling replaces field-level pooling
Machine tool insert	Cut metres	Spindle load / surface finish	Material batch, coolant, feed and speed programme	Same shape; part-to-part stagger replaces pit stagger
Rail wheel and brake	Axle kilometres	Profile drift / braking distance	Route gradient, load, weather	Same shape; regulatory interest in condition-based maintenance

source `src/tyremind/assets/profile.py` · `experiments/results/exp07_cross_domain.json`

> **What we deliberately do not claim**
>
> There is **no validated fleet result**, because no public dataset pairs tyre tread depth with vehicle telematics. We looked; it does not exist. The product shows fleet arithmetic clearly labelled as arithmetic, and says so on the screen rather than in a footnote. The rows below the two validated ones in the table above are architectural claims about problem shape, not measured results, and are labelled that way in the product too.

Why the transfer is a real technical claim and not a marketing one

The parts that transferred without modification were the latent state model (a level and a rate that drift over time), pooling a degradation baseline across a fleet of units, the uncertainty machinery, and remaining-life projection to a threshold. The parts that changed were vocabulary and configuration: "laps" became "flight cycles", "lap time" became a fused health index, and fuel and traffic were switched off because engines have neither.

That asymmetry is the point. If the transfer had required changing the model, the F1 model would have been a tyre model that happened to work. It required changing only the profile, which means the F1 model is a *degradation-under-confounding* model that happens to be pointed at tyres.

15 Scaling to production

What the current prototype can already carry, and what has to change

What already scales

Property	Measured	Why it holds at production volume
Per-lap live update	`0.22 ms mean, 0.74 ms max`	Cost is flat in session length by construction — a recursive filter carries a fixed-size state, so a lap on hour three costs what a lap on lap one costs. 921 laps measured.
Full session fit	`5.8 s`	Structured sparse transitions make propagation $O(n^2)$ rather than $O(n^3)$ in grid size. A full 20-car grid is already the tested case.
Strategy simulation	`50,000 races in 33 ms`	10,000 per option across five options; the interface uses 5,000 per option, which is 25,000 races in 17 ms. Fully vectorised with common random numbers across options, so comparisons are noise-free at any simulation count.
Offline operation	`8 sessions cached`	No network dependency at run time. The same property that makes a hackathon demo safe makes a trackside deployment safe.

What has to change, in order

	Change	Why it is needed	Approach
1	**Live timing feed adapter**	Today the live monitor replays a recorded session lap by lap. Production needs the real feed.	The estimator already consumes one lap at a time through a stable interface; only the source changes. The WebSocket layer and the filtered/smoothed separation are already built for it.
2	**Session state persistence**	A pit wall cannot lose its estimate to a process restart mid-race.	The filter state is a mean vector and a covariance matrix — a few hundred kilobytes. Checkpoint every lap; resume exactly, because the recursion is Markov in that state.
3	**Multi-tenant isolation**	Two teams must not share a cache, a fit, or a log line.	Per-tenant session store and fit cache. The estimator is stateless between fits, so isolation is a storage concern rather than a modelling one.
4	**Prior configuration per series**	The fuel and track priors are Formula 1 values. F2, GT3 and endurance have different fuel loads, burn rates and evolution profiles.	Promote `configs/physics.yaml` to a per-series profile with the same sensitivity experiment run per series, so each series ships with its own measured cost-if-wrong table.
5	**Hyperparameter warm starts**	The 5.8 s fit is dominated by L-BFGS-B over six log-variances from a cold start.	Seed from the previous session at the same circuit. Expected to cut fit time several-fold; the variance structure of a circuit is stable across a weekend.
6	**Model monitoring**	A degradation model that drifts is worse than none, because it is still confident.	Innovation z-scores are already computed. Production adds alerting on sustained same-signed innovations, plus tracking of practice-to-race bias per event so the +0.047 s/lap fig-

Change	Why it is needed	Approach
		ure is maintained rather than assumed constant.
7 **Backfill and benchmark suite**	Five events is a small validation set. The claim should be maintained against every event of a season.	Run the practice-to-race experiment across a full season nightly and publish the drift. This is a batch job over cached sessions, so it is a scheduling problem rather than a research one.

The scaling shape of the estimator itself

Cost is $O(n^2)$ per lap in the number of tracked states, where states are two per active car plus a small fixed set of shared coefficients. A twenty-car grid is roughly 125 states, and the measured 0.22 ms per lap leaves four orders of magnitude of headroom against a lap that takes eighty seconds. A hundred-car endurance field would be roughly twenty-five times the arithmetic and still comfortably real-time.

The binding constraint at scale is not compute. It is **data quality and prior calibration per series**, which is why items 4 and 7 above sit higher in the list than any performance work.

16 Commercial model and market

Where the money is, stated with the same discipline as the technical claims

Read this first

The figures in this section are **estimates and arithmetic, not measurements**. They are separated from the technical results deliberately, and the product labels them the same way on screen. Nothing here has the standing of section 08.

Why this is a product and not a paper

The estimator's output is a decision input, not a publication. Its value is realised at a specific moment — a strategist choosing a pit lap — and that moment recurs roughly twenty-four times a season per series, across dozens of series, for hundreds of teams. The unit of value is a decision made better under uncertainty, which is a recurring need rather than a one-off analysis.

Delivery model

Tier	For	What it includes	Rationale
Open core	Researchers, students, fans	The estimator, the synthetic benchmark, the experiment suite, the documentation corpus	The identifiability analysis is a contribution to the field and should be checkable. It is also the most credible marketing available.
Team licence	Junior formulae, customer teams	Live timing adapter, per-series prior calibration, session persistence, MCP endpoint, support	Per-series calibration is genuine recurring work and is the thing a team cannot easily do itself.
Enterprise	Works teams, manufacturers	On-premise deployment, prior replacement with team telemetry, custom asset profiles, joint validation against internal ground truth	A works team's value is the second opinion and the written-down assumptions, plus the ability to fold in data we do not have.
Industrial	Fleet, energy, aerospace	The same estimator under a different `AssetProfile`, with a domain validation programme	The C-MAPSS result is the proof that this is a real path rather than an aspiration — but each domain needs its own ground truth before a claim is made.

Market shape

Motorsport

Ten F1 teams, but hundreds of professional teams across F2, F3, IndyCar, WEC, IMSA, GT3, Formula E and national series — most with no tyre modelling capability and no realistic path to building one. This is a wide,

Adjacent industrial

Predictive maintenance is a large and well-funded category. What is scarce inside it is not modelling capacity but *honest uncertainty under confounding* — the exact property this method is built around.

Media

Broadcasters assert tyre degradation every Sunday without a number. A live, checkable figure with an interval is a graphics package, and it is the lowest-integration-cost entry point of the three.

underserved base, not a ten-customer market.

What better estimates are worth, as arithmetic

The product computes this from its own measured error rather than from a market assumption, and labels each figure by confidence. With naive error at 0.0966 s/lap and model error at 0.0044, the reduction in expected strategic regret follows from the degradation rate and the remaining stint length. On a twenty-lap remaining stint the difference between the two error levels is roughly two seconds of expected race time — approximately one championship position in a midfield fight, at roughly one race weekend in three.

We do not extrapolate that into a revenue figure. The honest statement is that the decision is worth more than the tool costs, and that the tool's error is measured rather than claimed.

17 Roadmap

From this prototype to a deployed system, with the research questions in the right order

NOW — prototype complete	0–3 months — one live series	3–9 months — multi-series	9–24 months — second industry
Estimator, physics, validation suite 9-view interface, 24 endpoints MCP server, hybrid RAG Offline demo, 326 tests C-MAPSS transfer validated	Live timing feed adapter Session-state checkpointing Per-series prior calibration Season-wide backfill of the practice-to-race claim Cross-session energy clock test	Multi-tenant isolation Warm-started hyperparameters Innovation-based drift alerting Prior replacement from team fuel telemetry Thermal state from IR where a series provides it	Second asset class with its own ground truth — battery or gearbox Per-domain sensitivity suite On-premise enterprise deployment Published identifiability analysis for degradation under confounding

The research questions, in the order they should be answered

1 — Does an energy clock absorb the +0.047 s/lap practice-to-race bias ACROSS sessions? Within a stint it provably cannot; across sessions it is untested and is the obvious next experiment.

2 — Does replacing the fuel prior with measured fuel mass narrow the interval as much as the sensitivity analysis predicts? This is the cheapest way to falsify our largest assumption.

3 — Does whole-field pooling still identify when the field is small? Halving the grid moved error 0.0047 → 0.0073; a ten-car junior series is the real test.

Figure 2 — Roadmap. The warm block is what exists today and has been measured. Everything to the right of it is engineering with a known shape, except the three research questions in the panel, which are stated as questions because they have not been answered.

What would falsify our central claim

Stated because a roadmap that cannot be wrong is a wish list. The central claim is that a physically-constrained state-space model attributes lap-time change to causes more accurately than the alternatives. It would be falsified by any of the following, and each is testable:

- A flexible learner recovering the hidden rate on the synthetic benchmark as accurately as the state-space model does. Today it cannot, because it has no such parameter — but a structured neural state-space model would have one.
- The practice-to-race bias failing to hold its sign across a full season, which would mean it is not the fuel-load effect we believe it is.
- Interval coverage falling below nominal on a larger real sample, which would mean the conservatism reported in section 11.2 is an artefact of the synthetic generator rather than a property of the estimator.
- A works team replacing our fuel prior with measured fuel and finding the estimate moves further than the sensitivity analysis predicts, which would mean the prior is doing more work than we claim.

18 References

Every source that shaped a design decision, and where in this document it appears

Reference	Used in	What it contributed
1 **Cappello, D. & Hoegh, A.** *A State-Space Approach to Modeling Tire Degradation in Formula 1.* arXiv:2512.00640, Nov 2025. arxiv.org/abs/2512.00640	§02, §03.3, §04, §09	The state-space framing; pit stops as resets; the single-driver limitation that motivates whole-field pooling.
2 **Pitwall: a calibrated race simulator with an LLM briefing layer.** arXiv:2607.06495, 2026. arxiv.org/abs/2607.06495	§02, §09, §13	Evidence that live calibrated simulation is practical; the discipline of gating generation on verified claims; their stated telemetry gap.
3 **Explainable Time Series Prediction of Tyre Energy in Formula One.** arXiv:2501.04067. arxiv.org/abs/2501.04067	§02, §06, §09	Tyre energy as the right intermediate quantity; explanation as first-class. Establishes the private-data limitation we route around.
4 **Rubber wear modelling: a review.** PMC12915245. pmc.ncbi.nlm.nih.gov/articles/PMC12915245	§06, §09	Archard's inadequacy for rubber; energy-based formulations as better supported; temperature belongs in the coefficient.
5 **Tyre wear model fusing viscoelasticity, roughness and thermodynamics.** Wear, S0043164824000565. sciencedirect.com/science/article/pii/S0043164824000565	§02, §06	The physical target, and the rig/FE data requirement that made an identifiable subset the correct scope.
6 **Heilmeier, A. et al.** Race simulation and pit-stop optimisation. github.com/TUMFTM/race-simulation	§02, §09	The reference race simulator. Establishes that a degradation rate is the key input taken as given — which is what we produce.
7 **Durbin, J. & Koopman, S.J.** *Time Series Analysis by State Space Methods*, 2nd ed., OUP 2012.	§05, §09	Sequential scalar updates (§6.4), the RTS smoother, and the prediction error decomposition that gives the exact likelihood.
8 **Rauch, H.E., Tung, F. & Striebel, C.T.** Maximum likelihood estimates of linear dynamic systems. *AIAA Journal*, 1965.	§05, §07	The smoother that produces the retrospective curve, kept strictly separate from the filtered live estimate.
9 **NASA Prognostics Center of Excellence.** C-MAPSS turbofan degradation simulation data set, FD001. nasa.gov/intelligent-systems-division — PCoE repository	§08.5, §14	Run-to-failure ground truth, the 125-cycle piecewise-linear RUL convention, and the asymmetric scoring function that penalises late predictions.
10 **Gneiting, T. & Raftery, A.E.** Strictly proper scoring rules, prediction, and estimation. *JASA*, 2007.	§08.3	CRPS as the scoring rule for probabilistic forecasts, which is why the model ladder scores distributions rather than point predictions.
11 **Cormack, G.V., Clarke, C.L.A. & Büttcher, S.** Reciprocal rank fusion outperforms Condorcet and individual rank learning methods. *SIGIR*, 2009.	§13	Rank fusion for the hybrid retrieval index. BM25 relevance and cosine similarity share no scale; ranks are comparable.
12 **Robertson, S. & Zaragoza, H.** The probabilistic relevance framework: BM25 and beyond. 2009.	§13	The lexical half of the retrieval index.
13 **Pacejka, H.B.** *Tyre and Vehicle Dynamics.* Butterworth-Heinemann.	§06, §11.4	The Magic Formula — surveyed and deliberately not implemented, because slip angle and slip ratio are not observable from public telemetry.

Reference	Used in	What it contributed
14 **Multilevel models of Formula 1 driver and constructor performance, 1950–2014.** Bristol / White Rose.	§09, §10	Establishes hierarchical modelling in motorsport, without applying it to degradation or using pit stagger as identification.
15 **FastF1.** Python package for F1 timing and telemetry. docs.fastf1.dev	§07	The data layer. Its event-resolution behaviour also produced one of the bugs recorded in §12.
16 **Model Context Protocol specification.** modelcontextprotocol.io	§13	The tool interface, and the practice guidance on tool count that fixed the server at seven.

Project documents

All committed to the repository and indexed by the product's own retrieval system, so any claim in this dossier can be traced to the file it came from.

- `docs/research/01_RESEARCH_AUDIT.md` — prior art, feasibility, what was cut and why
- `docs/research/03_DATA_AVAILABILITY.md` — what public F1 data contains and does not
- `docs/research/04_PHYSICS_FOUNDATION.md` — dynamics, thermal, wear, and their validation
- `docs/research/05_STATISTICAL_ARCHITECTURE.md` — the model and why a Kalman filter
- `docs/research/08_NOVELTY_ANALYSIS.md` — what is ours, what is not
- `docs/research/13_LIMITATIONS_AND_FAILURE_MODES.md` — where it is wrong
- `docs/model_card.md` — intended use, assumptions, performance
- `docs/INTEGRATIONS.md` — MCP and RAG, what each buys and what it does not

> **Closing**
>
> The brief asked for a model that strips external noise from practice sessions to produce clean degradation curves, and validates them against race-day pace. This dossier's argument is that the difficulty is not building such a model — it is *knowing whether you have*. Every number here exists to answer that second question, including the ones that do not flatter us.